

Data is all around us

INTRODUCTION TO DATA LITERACY



Jess Ahmet

Content Developer, DataCamp

	A	B	C	D	E	F	G	H
1	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category
2	768805383	Existing Customer	45	M	3	High School	Married	\$60K - \$80K
3	818770008	Existing Customer	49	F	5	Graduate	Single	Less than \$40K
4	713982108	Existing Customer	51	M	3	Graduate	Married	\$80K - \$120K
5	769911858	Existing Customer	40	F	4	High School	Unknown	Less than \$40K
6	709106358	Existing Customer	40	M	3	Uneducated	Married	\$60K - \$80K
7	713061558	Existing Customer	44	M	2	Graduate	Married	\$40K - \$60K
8	810347208	Existing Customer	51	M	4	Unknown	Married	\$120K +
9	818906208	Existing Customer	32	M	0	High School	Unknown	\$60K - \$80K
10	710930508	Existing Customer	37	M	3	Uneducated	Single	\$60K - \$80K
11	719661558	Existing Customer	48	M	2	Graduate	Single	\$80K - \$120K
12	708790833	Existing Customer	42	M	5	Uneducated	Unknown	\$120K +
13	710821833	Existing Customer	65	M	1	Unknown	Married	\$40K - \$60K
14	710599683	Existing Customer	56	M	1	College	Single	\$80K - \$120K
15	816082233	Existing Customer	35	M	3	Graduate	Unknown	\$60K - \$80K
16	712396908	Existing Customer	57	F	2	Graduate	Married	Less than \$40K
17	714885258	Existing Customer	44	M	4	Unknown	Unknown	\$80K - \$120K
18	709967358	Existing Customer	48	M	4	Post-Graduate	Single	\$80K - \$120K
19	753327333	Existing Customer	41	M	3	Unknown	Married	\$80K - \$120K
20	806160108	Existing Customer	61	M	1	High School	Married	\$40K - \$60K
21	709327383	Existing Customer	45	F	2	Graduate	Married	Unknown
22	806165208	Existing Customer	47	M	1	Doctorate	Divorced	\$60K - \$80K
23	708508758	Attrited Customer	62	F	0	Graduate	Married	Less than \$40K
24	784725333	Existing Customer	41	M	3	High School	Married	\$40K - \$60K
25	811604133	Existing Customer	47	F	4	Unknown	Single	Less than \$40K
26	789124683	Existing Customer	54	M	2	Unknown	Married	\$80K - \$120K
27	771071958	Existing Customer	41	F	3	Graduate	Single	Less than \$40K
28	720466383	Existing Customer	59	M	1	High School	Unknown	\$40K - \$60K
29	804424383	Existing Customer	63	M	1	Unknown	Married	\$60K - \$80K
30	718813833	Existing Customer	44	F	3	Uneducated	Single	Unknown
31	806624208	Existing Customer	47	M	4	High School	Married	\$40K - \$60K
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34	709029408	Existing Customer	41	M	4	Graduate	Married	\$60K - \$80K

Data comes in all shapes and sizes





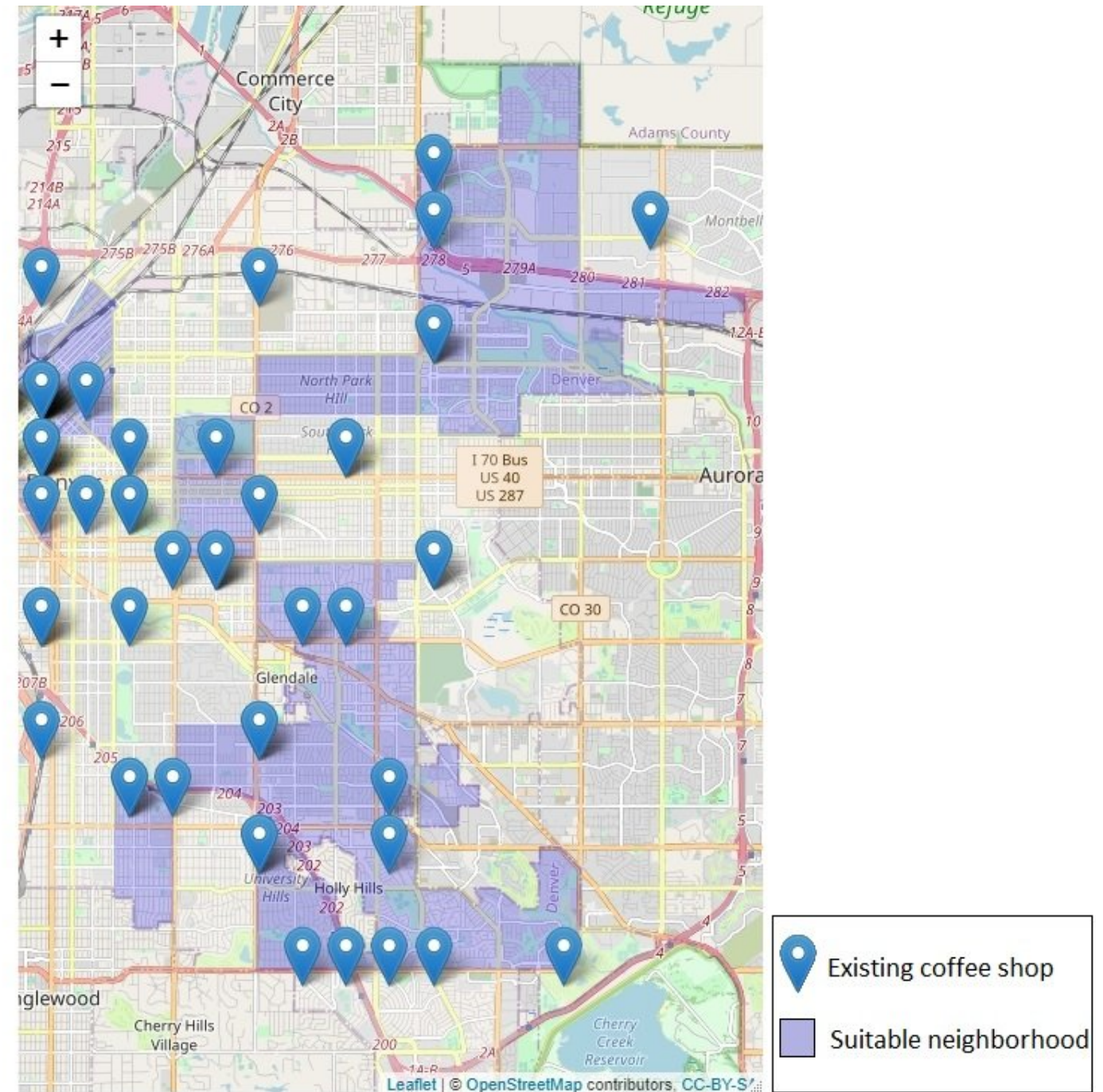
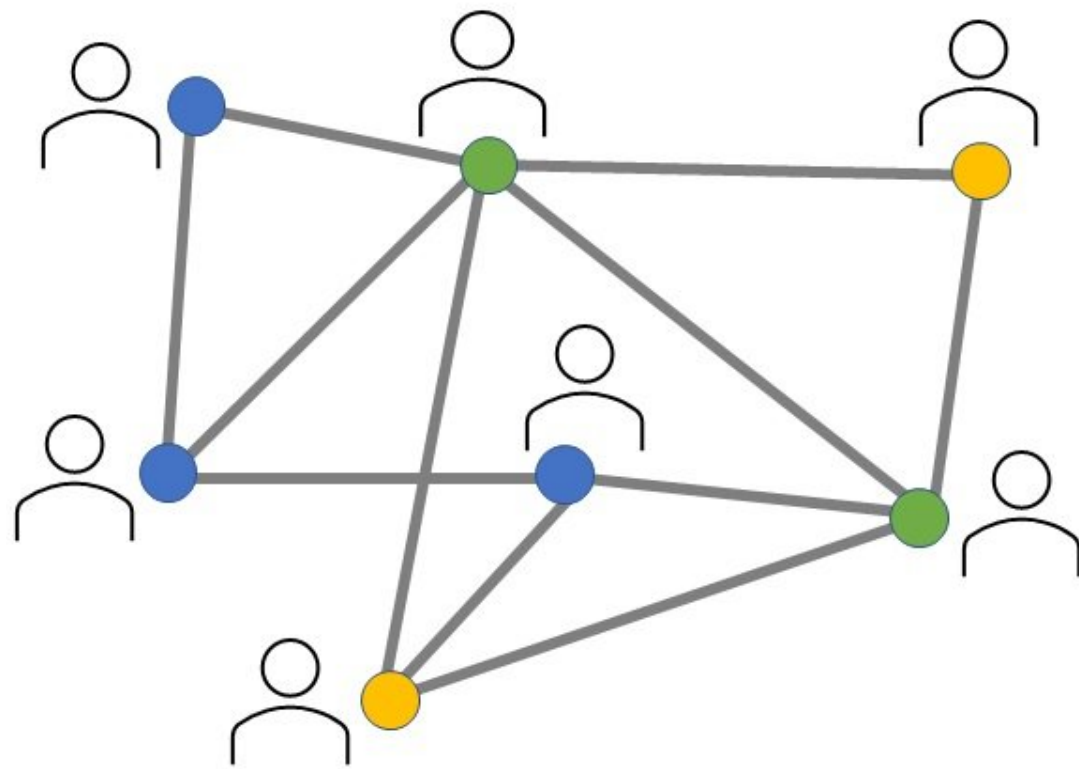
@DataCamp · 29 jul.

Happy Friday DataCampers! 🥳

What course will you be studying over the weekend?



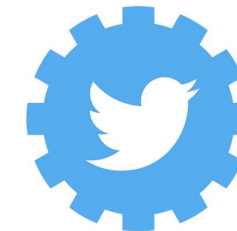
Data comes in all shapes and sizes



Data sources: open data

- Public databases and records
 - EU: data.europa.eu
 - US: data.gov
- Public APIs
 - Twitter
 - Google Maps
 - IMDB
 - Wikipedia
 - ...

data.europa.eu



 Google Maps Platform

Data sources: internal data

- Customer data
- Transaction data
- Website data
- Survey data



Let's practice!

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Common data types

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Importance of the data type

- Different types of data required to get a complete picture
- Has an effect on:
 - How to collect the data
 - How to store the data
 - How to analyze the data



Structured vs. unstructured data

Structured

- Data in tabular form
- Easy to search and organize
- Requires less preprocessing
- Stored in relational databases

Unstructured

- Data without a pre-defined structure
- Difficult to search and organize
- Requires more preprocessing
- Stored in document databases

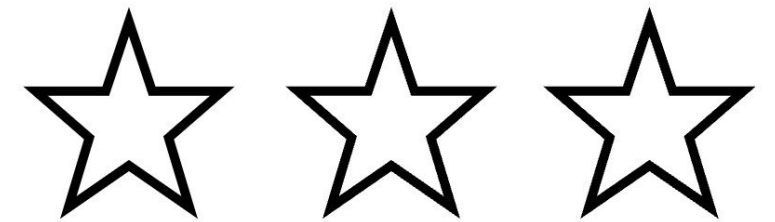
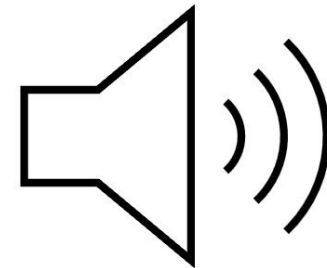
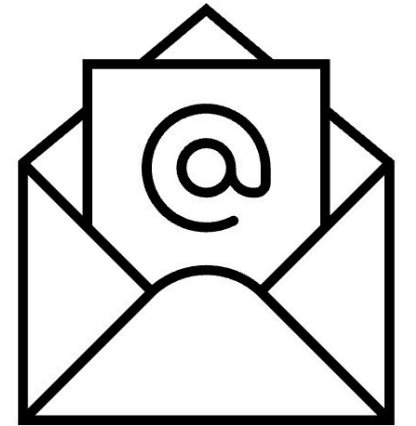
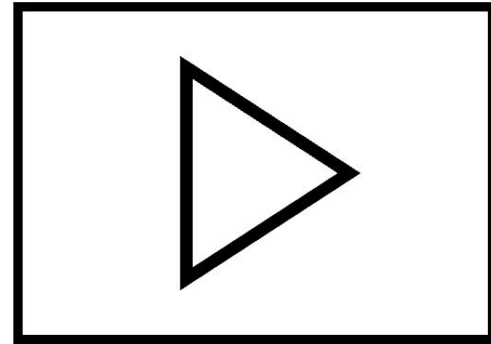
Examples of structured data

- Spreadsheets
- Data tables

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Examples of unstructured data

- Images
- Videos
- Sound
- Text



Quantitative vs. qualitative data

Quantitative

- Describes something with numbers
- Can be measured or counted
- Wider range of statistics and analysis methods
- Example: distance between home and work

Qualitative

- Describes something with categories
- Can be observed
- More restricted range of statistics and analysis methods
- Example: color of your eyes

Examples of quantitative data

- Count of ice cream scoops
- Minutes before the ice cream starts to melt
- Length of the cone



Examples of qualitative data

- Flavor of the ice cream
- Color of the ice cream
- Cone or cup?



Let's practice!

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Managing data

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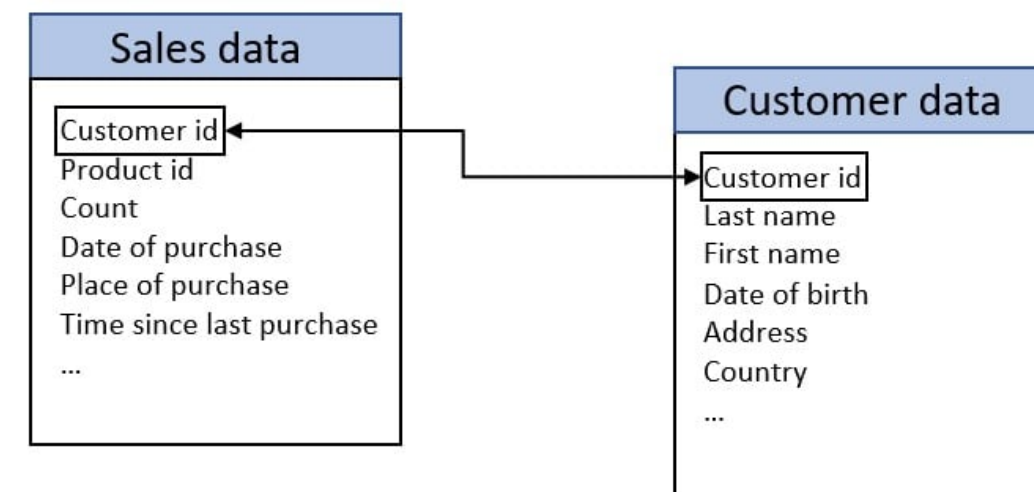
Databases: basic concepts

- Database management system (DBMS): software for managing databases
- Different types of databases:
 - Relational vs. document database
 - Data warehouse vs. data lake
- Designing and optimizing database systems is typically the responsibility of a data engineer

ORACLE



Relational database



Data storage in the cloud

- Data is stored on remote servers and accessed over the Internet
- Service provided by a specialized third party
- Cost-effective, but security dependent on third party



Google Cloud



Azure

Automation through data pipelines

- Pipelines move data from one database to another
- Automated collection and storage via the ETL process:
 - Extract
 - Transform
 - Load
- Ensures the availability of up-to-date and accurate data



Getting data from databases

- Querying: access and retrieve data
- Query with Structured Query Language (SQL)
- Further analysis with programming languages like R or Python



Dashboards

- Provides information at a glance, such as Key Performance Indicators (KPIs) and other metrics
- Receives data from a linked database
- Data is typically presented in a very visual
- Multipurpose tool: also used for exploratory analysis of the data and communicating
- Typically created and managed by a data analyst



Let's practice!

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Common data problems

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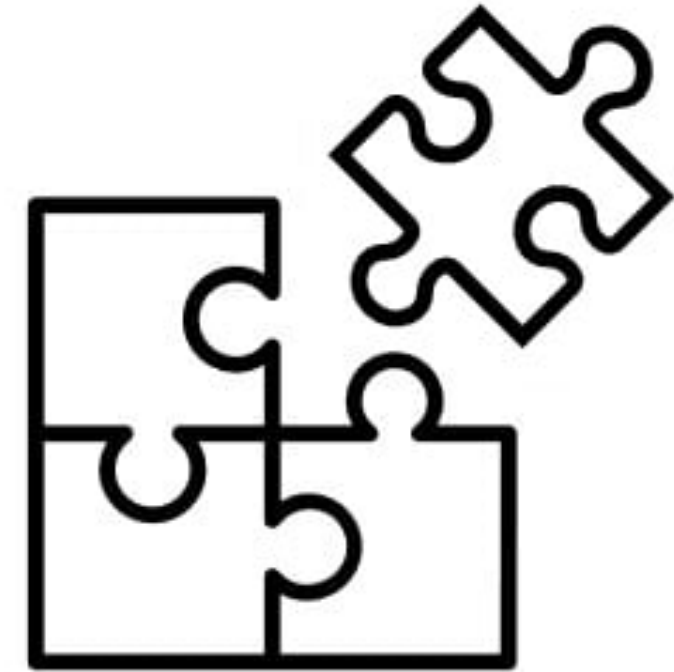
Dirty data

- Dirty data is:
 - Incorrect
 - Incomplete
 - Inconsistent
- Caused by human error, technical issues, or issues with the data collection process
- Garbage in, garbage out principle: dirty data can lead to wrong conclusions



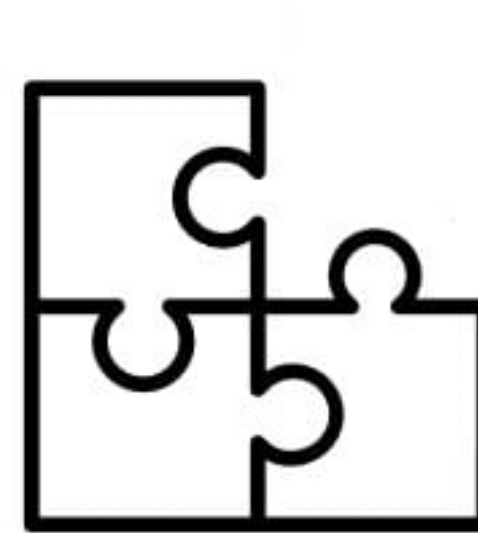
Data errors

- Data is incorrect or inconsistent
- Typically caused by human or technical error in recording the value or the format
- Techniques to counter:
 - If original value or valid format is known: correct data
 - If unknown: drop data



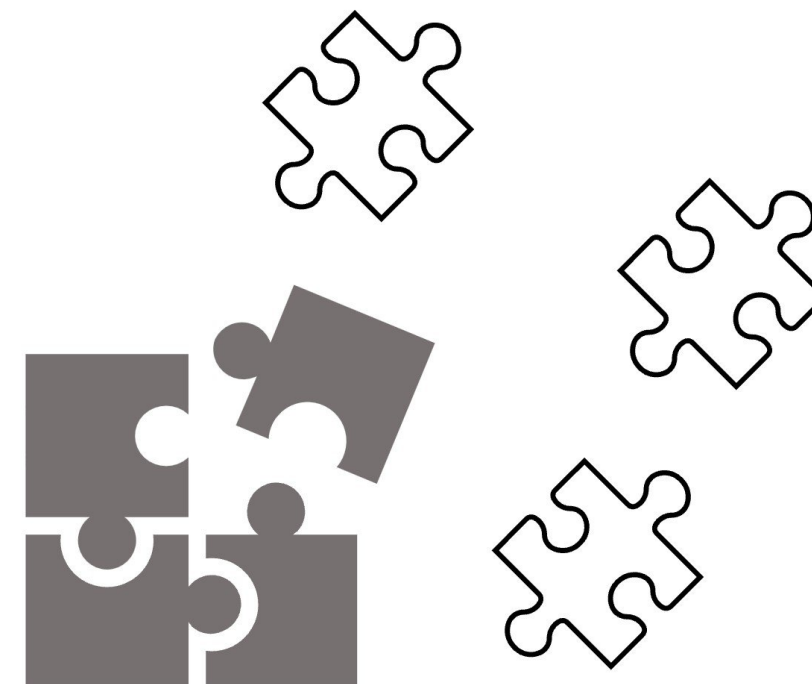
Missing data

- Data is incomplete
- Problematic if:
 - Many data points are missing
 - There are underlying patterns in the missing data
- Techniques to counter:
 - Dropping data
 - Imputation



Data bias

- Societal bias can be reflected in data as data bias
- Leads to unrepresentative data and therefore results
- Hard to detect and to resolve
- Techniques to counter:
 - Sound data collection process
 - Awareness in conclusions
 - Explainable AI models



Data cleaning

- Set of techniques to counter data problems
- Important preparation step for any data analysis
- But not all data problems are (completely) solvable
- It is always possible to do some kind of analysis



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